Data Structures (CST201) Dec 2023 Supplementary

Subject: Data Structures [CST201]

Curriculum Scheme: 2019 Scheme | Exam Year: 2023 | Type: SUPPLEMENTARY

Time: 3 Hours | Maximum Marks: 100 | Official Revision Paper

PART A - Answer all questions. Each carries 3 marks.

1. Define time and space complexity with respect to asymptotic notation.
2. Differentiate between Singly Linked List and Doubly Linked List with memory overhead.
3. State the condition for queue overflow and underflow in circular array implementation.
4. Explain properties of an AVL tree and illustrate Single and Double rotations.
5. What is hashing? Explain how separate chaining handles collision.

PART B - Module Questions. Answer one full question from each module.

6. (a) Write an algorithm to convert infix expression to postfix notation using stack. [8 marks]

(b) Evaluate the postfix expression: 12 4 / 5 * 3 + [6 marks]

7. (a) Construct a Binary Search Tree for elements: 50, 30, 70, 20, 40, 60, 80. [7 marks]

(b) Demonstrate deletion of a node with two children in a BST. [7 marks]

8. (a) Explain QuickSort partitioning algorithm and analyze worst-case time. [8 marks]

(b) Compare MergeSort and QuickSort on stability and memory efficiency. [6 marks]

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